



Development and characterization of microsatellite markers in *Castanea sativa* (Mill.)

Daniela Marinoni¹, Aziz Akkak¹, Giancarlo Bounous¹, Keith J. Edwards² and Roberto Botta^{1,*}

¹Dipartimento di Colture Arboree, Università di Torino, Via Leonardo da Vinci 44, Grugliasco, 10095, Torino, Italy; ²IACR Long Ashton Research Station, Department of Agricultural Science, University of Bristol, Long Ashton, BS41 9AF, Bristol, UK; *Author for correspondence (e-mail: roberto.botta@unito.it; fax: +39-011-6708658)

Received 23 April 2002; accepted in revised form 25 October 2002

Key words: Chestnut, Dinucleotide, DNA-typing, Polymorphism, SSR

Abstract

Thirty-three simple sequence repeat (SSR) markers were isolated and characterized in *Castanea sativa* (Mill.) from the cultivar Garrone Nero. For the identification of SSR loci, primers were designed on each side flanking the repeat region and they were initially tested on 5 chestnut samples using chemiluminescence detection. Twenty four loci were shown to be polymorphic and the number of alleles detected per locus varied from 2 to 7. Fourteen loci were chosen for the analysis of 20 cultivars grown in North Italy using the semi-automatic system ABI PRISM 377. These 14 markers showed a high level of genetic polymorphism with a total of 90 alleles; the number of alleles ranged from 4 to 10 per locus, with an average level of 6.4. The mean expected and observed heterozygosity were 0.724 (range: 0.649–0.835) and 0.793 (range: 0.350–0.950) respectively. The estimated frequency of null alleles showed a positive value for 3 loci, but except for 1 locus, the values were very low. The total value for the probability of identity was 7.04×10^{-11} . Paternity exclusion probability was very high (0.999), sufficiently high to study pollen flow.

Introduction

The European chestnut (*Castanea sativa*) is present in all countries along the Mediterranean basin, where climatic conditions are favourable; its forests extend from Caucasus through Turkey, Greece, Balkans to Italy, France, Spain, Portugal and South England.

The chestnut cultivation in Europe reached a peak in the 20th century at the beginning of the 1940's, with about 800.000 ha. After this period it began to reduce, both for the spread of fungal diseases (ink disease by *Phytophthora cinnamomi*, *Phytophthora cambivora* and canker blight by *Cryphonectria parasitica*), and for social-economic reasons (the depopulation of the hillside and mountain areas and the change of the food habits). Nevertheless since the beginning of the 1980's in all Europe there has been a marked turnaround with a considerable return of de-

mand in the first place for the nut and in the second place for the wood (Bounous et al. 2000). In the last thirty years the scientific community has as well become more interested in chestnut and many research programs have been promoted to rescue, collect and exploit the once cultivated germplasm. The genetic features of the various cultivars are in fact a precious resource as they are the result of centuries of work by man in the selection of the best tree types.

Traditional methods for cultivar identification are based on objective descriptions of tree and fruit characteristics, however, these observations are error-prone due to environmental variations affecting the expression of the genetic traits.

Biochemical markers, such as isozymes, are more consistent and they were used for studying genetic variability in chestnut (Dane et al. 1999; Fornari et al. 1999; Pereira et al. 1999; Villani et al. 1999).

Recently, the development of DNA markers has, however, provided a plenty of polymorphism to enable the rapid identification of cultivars and the construction of saturated genetic maps. Randomly amplified polymorphic DNA (RAPD) markers have been used for the characterization of chestnuts (Galderisi et al. 1998; Huang-Hong Wen et al. 1998; Mulcahy Bergamini et al. 1996; Santana et al. 1999), even if they are time and cost effective, they require an accurate standardization of the analysis to give reproducible patterns and they can be difficult to interpret due to their dominant mode of inheritance. In comparison, simple sequence repeats (SSR) have the potential to provide a more reliable method for DNA fingerprinting because of their high repeatability, high degree of polymorphism and co-dominant inheritance. SSR markers are abundant in most genomes and, generally, are uniformly distributed; they are PCR-based, thus requiring little DNA for the amplification, and every SSR locus is defined by a unique pair of primers, so that, information exchange between laboratories is easy and they are amenable to automation (Weber and May 1989; Yamamoto et al. 2002).

The use of multi-allelic, co-dominant markers for the analysis of heterozygous species is extremely useful since it allows individuals to be uniquely genotyped (Powell et al. 1996). This is particularly important for cultivar identification and to enhance the genetic characterization of germplasm collections and natural populations, permitting effective conservation of plant genetic resources; in addition molecular markers can be used for phylogenetic studies and linkage analysis.

In the present work we report the development and the characterization of 33 SSR markers in *Castanea sativa*. These markers are being used by the Department of Colture Arboree of Turin to characterize the most important cultivars of Piedmont Region and for studying the genetic structure of chestnut in five countries of Europe (Italy, France, Spain, United Kingdom, Greece) and the gene flow between populations at different domestication levels (orchard, copice, naturalized).

Material and methods

Plant material and DNA isolation

Twenty *Castanea sativa* individuals from the germplasm cultivated in Piedmont (North-west Italy) were used for testing the SSR loci. The 20 sampled cultivars were: Bracalla, Crou, Frattona, Gabiana, Garrone Nero, Garrone Rosso, Gentile, Giovasca, Lusençh, Madonna, Marrone of Chiusa Pesio, Marrubia, Neirana Exilles, Pelosa, Pelosa Grossa Cumiana, Pelosa Grossa Villar Pellice, Pelouset, Precoce di Brignola, Ruiana, Tempuriva.

Leaves and buds were harvested in spring or in winter respectively and reduced to a fine powder in liquid nitrogen using a mortar and pestle. DNA was extracted from leaves (approximately 0.5 g), following the procedure described by Thomas et al. (1993) in a TRIS-EDTA-NaCl buffer containing 0.25 M NaCl, 0.2 M Tris-HCl pH 7.6, 2.5% PVP 40.000, 0.05 M Na₂EDTA pH 8.0, 3% Sarcosyl, 20% ethanol and 1% β -mercaptoethanol.

In an alternative procedure, DNA was extracted from buds (approximately 0.1 g) according to Doyle and Doyle protocol (Doyle and Doyle 1990) with some modifications, in a 2% CTAB buffer (CTAB 2%, 1.4 M NaCl, 20 mM EDTA pH 8.0, 100 mM Tris-HCl pH 8.0, 1% PVP 40.000, 2% β -mercaptoethanol). In both cases the DNA was resuspended after purification in 70 μ l Tris-EDTA buffer.

A small progeny and its parent plants, used for a *Castanea sativa* mapping project (Casasoli et al. 2001), were analysed to check the segregation of alleles at each locus.

Isolation, cloning and sequencing of microsatellite repeats

The SSR library was prepared in collaboration with Long Ashton Research Station (UK) following the method by Edwards et al. (1996).

The approach chosen was to construct and screen a genomic library harboring small inserts (350–800 bp). The library was enriched with specific sequences in order to have a large set of positives clones and a large number of SSR types.

The eight following oligonucleotides were used to enrich for DNA fragments containing microsatellites: (GA)₁₅, (GT)₁₅, (CAA)₁₀, (CTG)₁₀, (CAG)₁₀, (CAT)₁₀, (ACT)₁₀, (GAC)₁₀. These oligonucleotides were prepared in 3X SSC (45 mM sodium citrate, pH

7.0 and 450 mM NaCl) and spotted onto a 0.5 cm² piece of Hybond N+ membrane and air-dried for 1 h. The dry membrane was UV-treated for 30 s using a 260-nm transilluminator. Weekly bound oligonucleotides were washed off the membrane by washing twice in 10 ml of hybridization buffer (50% formamide, 3X SSC, 25 mM Na-phosphate pH 7.0 and 0.5% SDS) at 45 °C for 2 days. The membranes were stored at -20 °C until use.

One µg of genomic DNA from *Castanea sativa* (cultivar Garrone Nero) was digested with *RsaI* for 2 h at 37 °C. Further, T4 ligase and 1 µg of *MluI* adapter (consisting of a 21-mer: 5'-CTCTTGCTTACGCGTGGACTA3' and a phosphorylated 25-mer: 5'-pTAGTCCACGCGTAAGCAAGAGCACAA3') were added to the mix and ligation was allowed to proceed for 2 h at 37 °C.

One µl of the ligation mix was amplified by PCR (Polymerase Chain Reaction) in a thermal cycler 9600 (Applied Biosystems), using the 21-mer primer at the following conditions: 25 cycles at 95 °C for 40 s, 60 °C for 60 s, 72 °C for 3 min with a final extension step of 10 min at 72 °C. After amplification, the sample was phenol-chloroform extracted, precipitated and resuspended in fresh sterile distilled water.

The enrichment for the microsatellites was carried out hybridizing for 2 days at 50 °C the ligated denatured DNA in 500 µl of hybridization buffer (50% formamide, 6X SSC, 25 mM NaH₂PO₄ pH 7.0, 2.5% SDS) containing 1 µg of the 21 mer oligonucleotide and the single Hybond N+ filter previously bound to the SSR oligonucleotides. Following hybridization, the filter was washed 5 times in 2X SSC, 0.01% SDS at 65 °C then 3 times in 0.5X SSC, 0.01% SDS at 65 °C. Finally, the hybridized DNA was eluted in 200 µl sterile distilled water by boiling for 5 min; furthermore, 1 µl of the eluted DNA was amplified by PCR using the 21-mer primer following these conditions: 25 cycles at 95 °C for 40 s, 60 °C for 60 s, 72 °C for 3 min with a final step of 72 °C for 10 min. After amplification, the PCR sample was phenol-chloroform extracted, precipitated and finally resuspended in 25 µl of fresh sterile distilled water.

Following this step, the enriched mix was digested with *MluI* and purified by Amersham Biosciences S-300 spin column according to the instructions supplied. The sample was vacuum dried and dissolved in sterile distilled water. One µl of the enriched DNA was then taken and ligated into 10 ng of a modified pUC 19 vector, containing a *Bss*HII site, for 16 h at room temperature.

DH5α competent cells (Invitrogen) were transformed with 2 µl of ligation mix, plated onto LB-agar containing ampicillin and incubated overnight at 37 °C. Following incubation, colonies were transferred into microtitre plates and stored until use.

At this stage of the research only dinucleotide repeats were selected for the isolation of SSR markers. The clones were transferred and replicated onto Hybond N+ membranes with a Biomek workstation (Beckman Instruments, Fullerton, CA, USA) to be hybridized with (CT)₁₀ and (CA)₁₀ radio-active probes for the detection of clones containing (GA)/(TC) and (CA)/(TG) repeats.

Eighty-five plasmids from individual positive colonies were extracted using Wizard *Plus* SV Minipreps DNA Purification System kit (Promega) and sequenced with a semi-automatic ABI-PRISM 377 sequencer.

Primers design

The identification of SSR loci requires characterization of unique sequences flanking each side of the repeat region to design locus-specific primers for PCR amplification of a given SSR.

The primers were designed so that the size of amplicons was kept between 90 and 300 bp to allow accurate sizing of PCR products. Primers were between 17 to 28 bases in length, with 35–60% GC content and matching or very close annealing temperature for primer pairs.

The rule of Suggs et al. (1981) was used for determining the annealing temperature that was further optimised experimentally for each primer pair.

SSR amplification and analysis

The primers were initially tested on 5 DNA samples (Garrone Rosso, Garrone Nero, Gentile, Marrone of Chiusa Pesio, Tempuriva) using chemiluminescence detection. PCR amplification of SSR loci was carried out using a reaction mixture (20 µl), consisting of 2 µl 10X buffer (100 mM Tris-HCl, pH 8.3, 500 mM KCl), 1.5 mM MgCl₂, 0.5 µM of each primer, 200 µM dNTP, 0.5 U *Taq*-DNA polymerase (AmpliTaq Gold polymerase, Applied-Biosystems) and 50 ng template DNA, were amplified in a MJ Research PTC 100 thermal cycler.

Amplification cycles consisted of an initial step of 9 min at 95 °C, followed by 28 cycles of 50 s at 94 °C, 45 s at the annealing temperature of each primer

pair, 90 s at 72 °C, with a final extension step of 7 min at 72 °C.

Two μl of each sample were analysed, after denaturation in 2 μl loading buffer (95% formamide, 20 mM EDTA, 0.05% bromophenol blue, 0.05% xylene cyanole) at 95 °C for 3 min on a 50 cm sequencing gel (6% polyacrylamide 24:1, 8 M Urea, 1X TBE). After electrophoresis, the DNA was blotted on a Zeta Probe (Bio-Rad, USA) positively charged nylon membrane and hybridized overnight in 15 ml hybridization buffer (0.5 M EDTA pH 8.0, 20% SDS, 1 M Na_2HPO_4 pH 7.2) containing 20 pmol of probe. The probes $(\text{GA})_{13}$ and $(\text{GT})_{13}$ were biotin labelled and hybridized at a temperature of 45 °C. Chemiluminescence detection was performed by the Phototope-Star Detection Kit (New England Biolabs, Beverly, MA, USA) following the instructions provided by the manufacturer.

Twenty samples were then analysed at 14 chosen loci using a semi-automatic system in order to evaluate the SSR markers polymorphism. In this case the forward primers were labelled with a fluorochrome (6-FAM, HEX or NED). One μl of a mix containing amplification products of three differently labelled loci, was added to 3 μl of a mix containing 10:2:1 parts of formamide, GeneScan-350 ROX size standard and loading buffer (25 mM EDTA, 50 mg mL^{-1} blue dextran). Samples were denaturated at 95 °C for 5 min and analysed on a sequencing gel (4.25% acrylamide, 1X TBE buffer, 6 M Urea) using a ABI-PRISM 377 DNA sequencer employing GeneScan software program (Applied Biosystems).

The analyses were repeated three times to show the reproducibility of the results.

Statistical analysis

The number and the frequency of alleles per locus, the expected and observed heterozygosity, the null alleles frequency, the paternity exclusion probability and the probability to find two identical genotypes were calculated with the program IDENTITY 1.0. (Wagner and Sefc 1999).

The expected heterozygosity (H_e) was calculated as $H_e = 1 - \sum p_i^2$ (Nei 1973) where p_i denotes the frequency of allele i in the cultivars. Observed heterozygosity (H_o) was obtained from direct calculation.

The null alleles frequency was calculated as $r = (H_e - H_o) / (1 + H_e)$, following Brookfield (Brookfield 1996), where H_e = expected heterozygosity and H_o = observed heterozygosity. The paternity exclusion

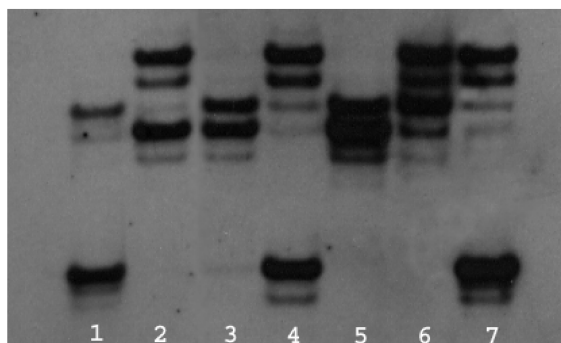


Figure 1. Mendelian inheritance at locus CsCAT14 in progeny plants used to test allele segregation; 1 and 2 are the parental genotypes, 3 to 7 progeny individuals

probability was calculated following Weir (Weir 1996) and the probability to find two identical genotypes was calculated as $PI = 1 - \sum p_i^4 + \sum \sum (2p_i p_j)^2$ (Paetkau et al. 1995) where p_i and p_j = frequency of alleles i and j respectively.

Results

A genomic library harboring small inserts (350–800 bp) was constructed from the cultivar Garrone Nero. The library was screened by hybridization with SSR probes $(\text{CT})_{10}$ and $(\text{CA})_{10}$. 18% of the genomic clones from the library $(\text{CT})_{10}$ and 28% from the library $(\text{CA})_{10}$ were positive.

Eighty-five positive clones were sequenced and 57 (67%) contained repetitive arrays with suitable flanking regions for primers design; the remaining 28 sequences were rejected for being redundant (7%), for lacking the SSR or due to the proximity of the microsatellite to the end of the insert.

All primer pairs were pre-screened on 5 individuals in a 6% polyacrylamide denaturing gel, using a chemiluminescence detection system. Forty-one primer pairs (72%) amplified at least one allele from all samples, whereas 16 primer pairs failed to amplify. Out of 41 primer pairs, 8 were discarded because produced genotypes with 3 or more alleles.

Allele segregation, tested in the small progeny (Figure 1), followed the Mendelian type of inheritance for all the loci.

The SSR of 18 (55%) out of the 33 selected loci were perfect types (Weber 1990) with 4 to 39 motifs (20 and 16 on average for CT and CA respectively); 12 (36%) were imperfect, 2 (6%) were compound and 1 (3%) was imperfect-compound.

Table 1. Primer sequences, repeat type and predicted product length of SSR loci isolated in *Castanea sativa* (cultivar: Garrone Nero); annealing temperature and number of alleles found analysing five chestnut cultivars (Garrone Rosso, Garrone Nero, Gentile, Marrone of Chiusa Pesio, Tempuriva). CsCAT = *Castanea sativa* Colture Arboree Torino

SSR Name		Primer sequence (5' → 3')	Repeat type	Predicted product length (bp)	Optimal Annealing temperature (°C)	No. alleles
CsCAT1	F	GAGAATGCCCACTTTTGCA	IMPERFECT	220	50°	6
	R	GCTCCCTTATGGTCTCG	(TG) ₅ TA(TG) ₂₄			
CsCAT2	F	GTAACCTGAAGCAGTGTGAAC	PERFECT	206	55°	7
	R	CGCATCATAGTGAGTGACAG	(AG) ₁₆			
CsCAT3	F	CACTATTTTATCATGGACGG	PERFECT	224	50°	4
	R	CGAATTGAGAGTTCATACTC	(AG) ₂₀			
CsCAT4	F	CATAGGTTCAAACCATACCCGTG	PERFECT	243	55°	3
	R	CTCATCTTTGTAGGGTATAATACC	(CA) ₂₃			
CsCAT5	F	CATTTTCTCATTGTGGCTGC	PERFECT	242	55°	2
	R	CATTGACATCCAATTAGG	(GA) ₂₀			
CsCAT6	F	AGTGCTCGTGGTCAGTGAG	IMPERFECT	180	50°	3
	R	CAACTCTGCATGATAAC	(AC) ₂₄ AT(AC) ₄			
CsCAT7	F	GAACATGATGATTGGCCTC	IMPERFECT	186	50°	3
	R	CCAAACATGACATATGTCCC	(TG) ₈ CG(TG) ₄			
CsCAT8	F	CTGCAAGACAAGAATTACAC	COMPOUND	210	50°	3
	R	GAATAACCTGCAGAAGGC	(GT) ₇ (GA) ₂₀			
CsCAT12	F	GGGTTGTATTAGGTATATCATGCC	PERFECT	304	60°	2
	R	CAGGATTTCACGGGAAAATCCG	(TG) ₁₉			
CsCAT14	F	CGAGGTTGTTGTTTCATCATTAC	PERFECT	164	58°	3
	R	GATCTCAAGTCAAAGGTGTC	(CA) ₂₂			
CsCAT15	F	TTCTGCGACCTCGAAACCGA	PERFECT	136	50°	2
	R	GCTAGGGTTTTCATTCTAG	(TC) ₁₂			
CsCAT16	F	CTCCTTGACTTTGAAGTTGC	PERFECT	143	50°	4
	R	CTGATCGAGAGTAATAAAG	(TC) ₂₀			
CsCAT17	F	TTGGCTATACTTGTCTGCAAG	IMPERFECT	147	58°	4
	R	GCCCCATGTTTTCTTCCATGG	(CA) ₁₉ A(CA) ₂ AA(CA) ₃			
CsCAT18	F	GCTTGATTGCACTGATAACC	IMPERFECT	108	58°	2
	R	CACATGAGCGCGTGCTCAGAAG	(CA) ₁₂ CG(CA) ₁₀			
CsCAT19	F	GTAAATGTATACCCACTAAACCGTC	IMPERFECT	180	60°	2
	R	GCTACTCAACCAATGATCCAAAC	(TG) ₁₀ CG(TG) ₁ TT(TG) ₄			
CsCAT20	F	GAATTAAGCAGTCCTCAATG	COMPOUND IMPERFECT	143	50°	2
	R	CAAGTCTCACCTTCAGTTAG	(AC) ₄ (AT) ₃ A(AT) ₃			
CsCAT21	F	GCCATACCTTTAGGACATTGAC	IMPERFECT	174	58°	2
	R	TCCTAAACATGAGAGTCCATC	(TG) ₁₀ TA(TG) ₁₂			
CsCAT22	F	GCTGTCTGATCTAGGATCAATC	PERFECT	240	54°	2
	R	CTAGCTTTAATATGAGCAAGGATGG	(AC) ₂₁			
CsCAT23	F	GCATGCTACTCAACCAAAGATCC	IMPERFECT	207	55°	1
	R	GATTGGGCCATCACAAAGTTTAC	(CA) ₂ CT(CA) ₄ CT(CA) ₁ CT(CA) ₃			
CsCAT24	F	AAACCGAAGATATAAGAGAGAG	PERFECT	214	50°	1
	R	CATTAGGCCAAATAGGCACC	(AG) ₁₇			
CsCAT25	F	CTCAATGGGACAGGCTTGATTTTG	COMPOUND	127	58°	1
	R	GTCTCACCTTCAGTTAGTCTGC	(AC) ₄ TT(TA) ₅			
CsCAT26	F	GAGACTTGAGATTGCAAAGG	PERFECT	116	50°	2

Table 1. Continued.

SSR Name		Primer sequence (5' → 3')	Repeat type	Predicted product length (bp)	Optimal Annealing temperature (°C)	No. alleles
	R	CTCACATTCAGTTAACAC	(CT) ₂₅			
CsCAT27	F	CGAATCTTTTGTGTGATCTAG	PERFECT	118	46°	1
	R	CTCCACTCAAAGGATTAAG	(AC) ₄			
CsCAT31	F	CCCTTTAAATACTGTGTGTG	IMPERFECT	91	45°	2
	R	CTACAGGAACACTCTGAATAG	(TG) ₄ T(TG) ₁₇			
CsCAT32	F	CAATTTTAAGCACAGGGATC	IMPERFECT	139	50°	1
	R	CAATGTGAATTGGCCATCAC	(AC) ₄ TC(AC) ₃			
CsCAT33	F	CTCGGAAACCAAACATGAATC	IMPERFECT	153	45°	1
	R	CGTTTTTGCTTCTTAGATTCC	(GA) ₅ GG(GA) ₁₁			
CsCAT34	F	TGAGCAAGGATGGATGATGAG	PERFECT	187	50°	4
	R	GGTGGTCATCATGACTGCATC	(GT) ₂₃			
CsCAT35	F	GAGAGGTGTCAAGAAGCTG	PERFECT	239	52°	1
	R	GTATGCTCACTAAACTGTCT	(AC) ₅			
CsCAT37	F	CAGTGAGCATGTAGTGTGCG	PERFECT	192	46°	2
	R	CACCTCTCTTCGTAATTTTCATG	(CA) ₁₂			
CsCAT38	F	CTAGAAATGAAAACCCCTAGC	PERFECT	160	48°	1
	R	CGTGAACCACGTATCTACC	(CT) ₇			
CsCAT39	F	TGAGTTCCTTAGCTTATCAAAGAGTGG	PERFECT	146	52°	2
	R	ACTAGGAATAGAGGCATACATCAATAGG	(AG) ₃₉			
CsCAT40	F	CCCTTGATAATTCTGTTCTG	IMPERFECT	198	50°	1
	R	GTCGCGGCATATTACTACC	(GT) ₆ G(GT) ₃			
CsCAT41	F	AAGTCAGCAACACCATATGC	PERFECT	224	50°	3
	R	CCCCTGTTCATGAGTTTCT	(AG) ₂₀			

Twenty-four (73%) loci out of 33 resulted polymorphic among the 5 individuals. The number of alleles detected per locus varied from 2 to 7: 12 loci showed 2 alleles, 6 loci 3 alleles, 4 loci 4 alleles and the remaining 2 loci 6 and 7 alleles respectively (Table 1).

Fourteen (CsCAT1, CsCAT2, CsCAT3, CsCAT4, CsCAT5, CsCAT6, CsCAT7, CsCAT8, CsCAT14, CsCAT15, CsCAT16, CsCAT17, CsCAT34, CsCAT41) out of the 28 loci were chosen for the analysis on the ABI PRISM 377 DNA sequencer on the basis of the polymorphism shown in the first stage of the work.

The twenty DNA samples extracted from cultivars of *Castanea sativa* grown in Piedmont Region were amplified at the 14 loci and analysed on ABI PRISM 377. They showed at least 1 or 2 alleles for each locus.

The number of alleles found was 90 in total and ranged from 4 to 10 per locus (Table 2), with an average of 6.4. The loci with higher allele number were

CsCAT3 (10 alleles) and CsCAT41 (9 alleles). The largest size range between alleles was 79 bp in CsCAT3.

Statistical elaboration of data is reported in Tables 2 and 3. The higher allele frequency was 0.525 for allele 145 at locus CsCAT16, whereas other alleles were specific for a particular cultivar with allele frequency of 0.025.

The mean expected heterozygosity (He) was 0.724 (range: 0.649–0.835). The mean observed heterozygosity (Ho) was 0.793 (range: 0.350–0.950). The locus CsCAT41 showed the largest difference with He = 0.835 and Ho = 0.950.

The estimated frequency of null alleles showed positive value for 3 loci among 14 SSR markers and precisely for CsCAT2 (0.011), CsCAT3 (0.002), CsCAT34 (0.181).

The total paternity exclusion probability was 0.9999; this index was high for CsCAT2 (0.650) and

Table 2. Allele size and allele frequency of 14 SSR loci studied in 20 chestnuts cultivars

LOCUS	ALLELE SIZE (bp)	ALLELE FREQUENCY	LOCUS	ALLELE SIZE (bp)	ALLELE FREQUENCY
CsCAT1-NED	194	0.250	CsCAT8-NED	200	0.375
	207	0.150		202	0.125
	209	0.150		204	0.100
	215	0.125		210	0.200
	223	0.325		214	0.200
CsCAT2-HEX	207	0.050	CsCAT14-6-FAM	135	0.375
	211	0.100		144	0.075
	213	0.275		152	0.400
	215	0.075		163	0.150
	219	0.050	CsCAT15-HEX	125	0.050
	225	0.250		127	0.325
	229	0.125		133	0.025
	231	0.075		137	0.425
CsCAT3-6-FAM	190	0.050		139	0.050
	225	0.500		143	0.050
	227	0.050		160	0.075
	233	0.050	CsCAT16-HEX	127	0.100
	235	0.025		132	0.125
	241	0.175		134	0.100
	251	0.025		143	0.025
	253	0.025		145	0.525
	257	0.075		147	0.100
	269	0.025		157	0.025
CsCAT4-6-FAM	215	0.175	CsCAT17-6-FAM	140	0.050
	223	0.275		142	0.225
	227	0.025		147	0.125
	237	0.025		151	0.500
	239	0.400		156	0.075
	241	0.100		163	0.025
CsCAT5-HEX	216	0.250	CsCAT34-HEX	146	0.075
	231	0.025		155	0.400
	233	0.300		166	0.050
	239	0.150		169	0.425
	245	0.275		189	0.050
CsCAT6-6-FAM	161	0.325	CsCAT41-NED	205	0.250
	167	0.025		215	0.150
	175	0.325		220	0.025
	181	0.125		223	0.025
	185	0.050		225	0.050
	187	0.025		227	0.100
	195	0.025		231	0.200
	197	0.100		237	0.150
CsCAT7-HEX	188	0.500		239	0.075
	208	0.150			
	212	0.025			
	214	0.050			
	224	0.275			

Table 3. Expected and observed heterozygosity, estimated frequency of null alleles, paternity exclusion probability and probability of identity of 14 SSR loci studied in 20 chestnut cultivars

LOCUS	Heterozygosity Expected (He) and Observed (Ho)	Frequency of Null alleles	Paternity Exclusion Probability	Probability of Identity
CsCAT1-NED	He: 0.771 Ho: 0.950	−0.101	0.555	0.160
CsCAT2-HEX	He: 0.820 Ho: 0.800	0.011	0.650	0.099
CsCAT3-6-FAM	He: 0.704 Ho: 0.700	0.002	0.512	0.160
CsCAT4-6-FAM	He: 0.722 Ho: 0.750	−0.016	0.489	0.211
CsCAT5-HEX	He: 0.748 Ho: 0.850	−0.058	0.510	0.198
CsCAT6-6-FAM	He: 0.759 Ho: 0.900	−0.080	0.548	0.164
CsCAT7-HEX	He: 0.649 Ho: 0.650	−0.0006	0.370	0.287
CsCAT8-NED	He: 0.754 Ho: 0.950	−0.112	0.533	0.172
CsCAT14-6-FAM	He: 0.671 Ho: 0.900	−0.137	0.404	0.294
CsCAT15-HEX	He: 0.700 Ho: 0.800	−0.059	0.468	0.229
CsCAT16-HEX	He: 0.677 Ho: 0.800	−0.073	0.475	0.186
CsCAT17-6-FAM	He: 0.675 Ho: 0.750	−0.045	0.448	0.226
CsCAT34-HEX	He: 0.649 Ho: 0.350	0.181	0.385	0.319
CsCAT41-NED	He: 0.835 Ho: 0.950	−0.063	0.672	0.089

yPfor CsCAT41 (0.672), with a mean value of 0.474 (range: 0.385–0.672).

The probability of identity (PI) changed from 0.089 for CsCAT41 to 0.319 for CsCAT34, the mean value was 0.177. According to frequency results the total probability of identity was 7.04×10^{-11}

Discussion

The most frequently occurring microsatellite dinucleotide repeats in plants are (AT) with (CT)/(AG) and (CA)/(TG) as the second and third most frequent ones (Lagercrantz et al. 1993; Thomas and Scott 1993; Wang et al. 1994). However due to the difficulties in screening for (AT), autocomplementary oligonucleotides, using hybridization techniques, we decided to

develop microsatellite markers containing (CT)/(AG) and (CA)/(TG) repeats. We found that there was a prevalence of (CA)/(TG) positive clones (41.4%) on (CT)/(AG) clones (22.4%).

From the initial screening of 85 positive clones we designed 57 (67%) primer pairs; this value is comparable with data (72%) by Gianfranceschi et al. (1998).

Thirty-three out of the tested 57 primer pairs amplified an expected number of alleles per genotype, 8 were discarded because they showed extra bands and 16 failed to amplify probably due to mismatches in the annealing region or to artifact inserts originated during the enrichment procedure.

Only 6% (2 clones) of the 33 microsatellite sequences contained a compound repeat type. The observed number is in agreement with other studies; in fact only 7% of di- and trinucleotide soybean repeats

Äare compound (Akkaya et al. 1995) while none was found during development of 30 Arabidopsis dinucleotide markers (Bell and Ecker 1994).

From the average length of (CT)/(AG) and (CA)/(TG) motifs we expected a high polymorphism, as asserted by Weber (1990); among 33 loci, 24 (73%) were polymorphic after chemiluminescence detection. We didn't observe polymorphism for loci with less than 5–8 repeats as it was observed and demonstrated in human SSR by Armour et al. (1994), Valdes et al. (1993) and Weber (1990).

In order to define a set of microsatellite sequences useful for the fingerprinting and mapping of chestnut, we choose 14 loci for the analysis of 20 cultivars. At first, loci with 3 or more alleles, detected using chemiluminescence, were selected; later also some loci with only 2 alleles were considered, to check if their polymorphism could be useful for studying a larger number of samples; indeed locus CsCAT5 showed 5 alleles and CsCAT15 showed 7 alleles; so also loci with 2 alleles at the first step of analysis may be useful for fingerprinting.

The polymorphic level was good, considering that we analysed 20 genotypes from a small geographic area. This fact reduced the probability of finding a higher allelic diversity which can be expected when further individuals will be studied. Most of the 14 loci can be rated as highly informative, also for the value of heterozygosity.

The occurrence of null alleles was already pointed out by Callen et al. (1993) as a possible problem associated with the use of microsatellite markers. If undetected, a null allele will merely result in that individual being scored as a homozygote resulting in loss of information. The presence of a null allele at an appreciable frequency can be suspected (within a homogeneous population) when the observed heterozygosity is markedly less than the expected heterozygosity. For 2 loci (CsCAT2, CsCAT3) a positive value for null allele frequency was found but was very low, and thus not giving conclusive indication about the issue; only 1 locus (CsCAT34) showed a high value of null allele frequency (0.181) and can be regarded as a locus with a possible occurrence of null alleles, although homozygote excess may not always be obviously attributable to the presence of non amplifying alleles (O'Reilly and Wright 1995). The negative value for the null allele index, found in most loci, on the contrary, could associate the homozygotes with unfavourable characteristics, eliminated by man selection of cultivated germplasm.

The ability of genetic markers to study pollen flow is represented by the paternity exclusion probability, which is the parameter used to describe the chance of correctly identifying pollen donors (Tanaka et al. 1999). This value was particularly high at loci CsCAT41 (0.672) and CsCAT2 (0.650) respectively and the very high combined probability of paternity exclusion (0.9999) also indicates that this set of markers will be useful in parentage studies even in situations when both parental individuals are unknown.

The range of variability makes it possible to select loci for specific applications. For example markers with only a few alleles are well-suited for population genetic studies (Carvalho and Hauser 1994). The more variable microsatellite loci however are ideal for determination of parentage, pedigree analysis and for genome mapping (O'Reilly and Wright 1995). Some of the studied loci are being used to construct a genetic linkage map of European chestnut; preliminary results showed that CsCAT1 and CsCAT15 are tightly linked (Barreneche et al. 2002).

In this study we described 33 SSR markers developed for *Castanea sativa*. We began to evaluate in details 14 markers; among these 14 loci, 8 are particularly interesting considering their good polymorphism and are being used within the EU CASCADE project for the study of gene flow and genetic mapping (Barreneche et al. 2002).

Acknowledgements

Authors wish to thank Dr. Fiorella Villani for providing the DNA of the progeny. The research was funded by the Regione Piemonte Administration and by the EU contract EVK2 – CT – 1999-00006, CASCADE project.

References

- Akkaya M.S., Shoemaker R.C., Specht J.E., Bhagwat A.A. and Cregan P.B. 1995. Integration of simple sequence repeat DNA markers into a soybean linkage map. *Crop Sci.* 35: 11439–11445.
- Armour J.A.L., Neumann R., Gobert S. and Jeffreys A.J. 1994. Isolation of human simple repeat loci by hybridization selection. *Hum. Mol. Genet.* 3: 599–605.

- Barreneche T., Casasoli M., Russell K., Akkak A., Villani F. and Kremer A. 2002. Comparative genetic mapping between *Quercus robur* L. and *Castanea sativa* Mill. Plant Animal & Microbe Genomes X January: 12–16.
- Bell C.J. and Ecker J.R. 1994. Assignment of 30 microsatellite loci to the linkage map of *Arabidopsis*. Genomics 19: 137–144.
- Bounous G., Botta R. and Beccaro G. 2000. The chestnut: the ultimate energy source. Nutritional value and alimentary benefits. NUCIS newsletter 9: 44–50.
- Brookfield J.F.Y. 1996. A simple new method for estimating null allele frequency from heterozygote deficiency. Mol. Ecol. 5: 453–455.
- Callen D.F., Thompson A.D., Shen Y., Philips H.A., Richards R.I., Mulley J.C. et al. 1993. Incidence and origin of "null" alleles in the (AC)_n microsatellite markers. Am. J. Hum. Genet. 52: 922–927.
- Carvalho G.R. and Hauser L. 1994. Molecular genetics and the stock concept in fisheries. Reviews in Fish Biology and Fisheries 4: 326–350.
- Casasoli M., Mattioni C., Cherubini M. and Villani F. 2001. A genetic linkage map of European chestnut (*Castanea sativa* Mill.) based on RAPD, ISSR and isozyme markers. Theor. Appl. Genet. 102: 1190–1199.
- Dane F., Hawkins L.K., Huang-Hong Wen and Huang H.W. 1999. Genetic variation and population structure of *Castanea pumila* var. *ozarkensis*. J. Am. Soc. Hort. Sci. 124: 666–670.
- Doyle J.J. and Doyle J.L. 1990. Isolation of plant DNA from fresh tissue. Focus 12: 13–15.
- Edwards K.J., Barker J.H.A., Daly A., Jones C. and Karp A. 1996. Microsatellite libraries enriched for several microsatellite sequences in plants. Biotech. 20: 758–760.
- Fornari B., Turchini D. and Villani F. 1999. Genetic structure and diversity of two Turkish *Castanea sativa* Mill. populations investigated with isozyme and RAPD polymorphism. Journal of Genetics and Breeding 53: 315–325.
- Galderisi U., Cipollaro M., Di Bernardo G., De Masi L., Galano G. and Cascino A. 1998. Molecular typing of Italian sweet chestnut cultivars by random amplified polymorphic DNA analysis. J. Hort. Sci. Biotech. 73: 259–263.
- Gianfranceschi L., Seglias N., Tarchini R., Komjanc M. and Gessler C. 1998. Simple Sequence Repeats for the genetic analysis of apple. Theor. Appl. Genet. 96: 1069–1076.
- Huang-Hong Wen, Dane F., Kubisiak T.L. and Huang H.W. 1998. Allozyme and RAPD analysis of the genetic diversity and geographic variation in wild populations of the American chestnut (Fagaceae). Am. J. Bot. 85: 1013–1021.
- Lagercrantz U., Ellegren H. and Anderson L. 1993. The abundance of various polymorphic microsatellite motifs differ between plants and animals. Nucl. Acid Res. 21: 1111–1115.
- Mulcahy Bergamini G., Trautweiler A., Linskens H.F. and Mulcahy D.L. 1996. Genetic Characterization of Chestnut. In: Linsken H.F. and Jackson J.F. (eds), Modern Methods of Plant Analysis. Vol. 18., pp. 149–154.
- Nei M. 1973. Analysis of gene diversity in subdivided populations. Proc. Nat. Acad. Sci. USA 70: 3321–3323.
- O'Reilly P. and Wright J.M. 1995. The evolving technology of DNA fingerprinting and its application to fisheries and aquaculture. Journal of Fish Biology 47: 29–55.
- Paetkau D., Calvert W., Stirling I. and Strobeck C. 1995. Microsatellite analysis of population structure in Canadian Polar bears. Mol. Ecol. 4: 347–354.
- Pereira M.J.P., Castro L.F.T., Torres-Pereira J.M.G. and Lorenzo S.P. 1999. Isozyme polymorphism in portuguese chestnut cultivars. Acta. Hort. 494: 283–286.
- Powell W., Macharay G.C. and Provan J. 1996. Polymorphism revealed by simple sequence repeats. Trends Plant Sci. 1: 215–222.
- Santana C., Oliveira M. and Valdivieso T. 1999. Molecular typing of rootstock hybrids (*Castanea sativa* × *Castanea crenata*) and portuguese *Castanea sativa* cultivars based on RAPD markers. Acta Hort 494: 295–301.
- Suggs S.V., Wallace R.B., Hirose T., Kawashima E.H. and Itakura K. 1981. Use of synthetic oligonucleotides as hybridisation probes: Isolation of cloned cDNA sequences for human β_2 -microglobulin. Proc. Nat Acad. Sci. 78: 6113.
- Tanaka K., Tsumura Y. and Nakamura T. 1999. Development and polymorphism of microsatellite markers for *Fagus crenata* and the closely related species, *F. japonica*. Theor. Appl. Genet. 99: 11–15.
- Thomas M.R., Matsumoto S., Cain P. and Scott N.S. 1993. Repetitive DNA of grapevine: classes present and sequences suitable for cultivar identification. Theor. Appl. Genet. 86: 985–990.
- Thomas M.R. and Scott N.S. 1993. Microsatellite repeats in grapevine reveal DNA polymorphism when analysed as sequence tagged sites (STSs). Theor. Appl. Genet. 86: 985–990.
- Valdes A.M., Slatkin M. and Freimer N.B. 1993. Allele frequencies at microsatellite loci: the stepwise mutation model revisited. Genetics 133: 737–749.
- Villani F., Lauteri M., Sansotta A., Cherubini M., Monteverdi M.C., Mattioni C. et al. 1999. Genetic structure and quantitative traits variation in F1 full-sibs progenies of *Castanea sativa* Mill. Acta Hort. 494: 395–405.
- Wagner H.W. and Sefc K.M. 1999. IDENTITY 1.0. Centre for Applied genetics, University of Agricultural Sciences, Vienna.
- Wang Z., Weber J.L., Zhong G. and Tanksley S.D. 1994. Survey of plant short tandem DNA repeats. Theor. Appl. Genet. 88: 1–6.
- Weber J.L. 1990. Informativeness of human (dC-dA)_n (dG-dT)_n polymorphisms. Genomics 7: 524–530.
- Weber J.K. and May P.E. 1989. Abundant class of human DNA polymorphism which can be typed using the polymerase chain reaction. Am. J. Hum. Genet. 44: 388–397.
- Weir B.S. 1996. Genetic Data Analysis II. Sinauer Assoc, Sunderland, Mass, USA.
- Yamamoto T., Kimura T., Sawamura Y., Manabe T., Kotobuki K., Hayashi T. et al. 2002. Simple sequence repeat for genetic analysis in pear. Euphytica 124: 129–137.